The idea of this research project was to use machine learning to investigate in how far the performance in terms (e.g. gas adsorption (standard is nitrogen, carbon dioxide, methane, or heat capacity) of metal-organic frameworks is determined by the atomic composition of the materials (made in a lego-style fashion using organic lingers that are attached to metal centers), and in how far it is determined by the underlying crystallographic net.

I did not get the performance parameter from my collaborators; I don't know whether they ever started calculating them. The databases I took the structures from are the CCDC, CoreMof, and CSD Mof collection. These are very standard, and surely their performance parameter in terms of gas adsorption must be somewhere online. E.g., for the QMOF database, which has overlap with the ones I used, they can be found here https://archive.materialscloud.org/record/2022.101.
The heat capacity of the materials in the CoreMof database can be found here https://archive.materialscloud.org/record/2022.53

(I never used that, this just showed up in the depository)

I am sharing data in two file formats .ado and .dat, both can be opened with standard text editors.

The ado files contain the analysis and description of the crystallographic net types. The dat files contain the names of the materials to match with database containing performance properties, and the chemical compositions of the building blocks.

The materials are taken from three different databases named databasename = CCDC, CSD, CoreMOF. There might be overlaps in the databases.

Since the "crystallographic net" of a material is not well defined, I computed five different net constructions, once including and once excluding metal-metal bonds (all shared). The best one to start with is the single-node simplification without metal-metal interactions, that is the files databasename_noMeMe_no01_SingleNodeSimpl_no012. There, take always the one without "specific" in the name, and only for materials that do not have an entry in that database, take the entry from the corresponding "specific" file instead. Furthermore, there are many structures with several entries in the ado files. You can use all of them together as a descriptor for the material, but for a start I would suggest to just take the one with the smallest number in the name, i.e. the smallest number behind "intercluster bonds for rings".

The materials have names such as ABUWOJ. The files with the net information (ado format) **can be matched** with the structure names using the dat.format files. The entry “3" in the left file is the third structure of the SSDS database, it is called “ABUWOJ" as stated in the “Comment" in the dat file.

CCDC_noMeMe_no01_SingleNodeSimpl_no012.ado

```
##################################################################
3:(C32 H38 O18 S4 Zn4)n,2n(H2 O1)/intercluster bonds for rings>6
##################################################################
3D framework with ZA
There are 2 interpenetrating nets
FISE: Full interpenetration symmetry elements
-----------------------------------------------
1: 2(1)[0,0,1]
2: 2(1)[1,0,0]
-----------------------------------------------
PIC: [1,0,0][0,1,0][0,0,1] (PICVR=1)

Zt=1; Zn=2

Class IIa  Z=2

Coordination sequences
----------------------
ZA1:  1  2  3   4   5   6   7   8    9   10
Num   6 18 38  66 102 146 198 258  326  402
Cum   7 25 63 129 231 377 575 833 1159 1561
----------------------
TD10=1561

Vertex symbols for selected sublattice
--------------------------------------
ZA1 Point symbol:{4^12.6^3}
Extended point symbol:[4.4.4.4.4.4.4.4.4.4.4.4.6(4).6(4).6(4)]
--------------------------------------
Point symbol for net: {4^12.6^3}
```

CCDC_noMeMe_no01_SingleNodeSimpl_no012.dat

```
3:(C32 H38 O18 S4 Zn4)n,2n(H2 O1)/intercluster bonds for rings>6
Name: C32 H42 O20 S4 Zn4. catena-(bis(\m~4~-1,4-Benzenedicarboxylate)-(\m~2~-1,4-
benzenedicarboxylate)-bis(\m~3~-hydroxo)-tetrakis(dimethyl sulfoxide)-tetrazinc dihydrate).
Author(s):
Journal:  Year: 0 Volume:  Number:  Pages:
List of atoms
Name Num DegOx  Wyckoff    X        Y        Z     AtCode       S     CN         Rsd
ZA    1          4a,Ci   0.50000  0.00000  0.50000    1         <  6>
Composition: ZA
CCF:
Space Group: P b c a; Mult.= 8
Crystal Class: D2h
Z= 4
Pearson Symbol: oP4
Wyckoff sequence: a
Standard Wyckoff: a
Other Wyckoff sequences:
b
ABC Formula: A4
Cell Parameters
A= 14.4988  B= 17.1659  C= 18.1536
Alpha= 90.000  Beta= 90.000  Gamma= 90.000
Volume=4518.16; D(ter)= 0.001
Comment: File: ABUWOJ_CCDC.cif.
CENTAT: none
SIMPLIG: ZA=C4O10Zn4 ZB=C6 ZC=C8O2
SIMPPAR: METH=Stand; INTRA=V; INTER=H
-----------------------------------------------
```

The chemical composition is given in the title, e.g. C32 H38 O18 S4 Zn4. The chemical composition of the metal nodes and the organic ligands is recorded in the dat files in the CENTAT and the SIMPLIG lines. In the example above there is no central atom but three atom sets C4O10Zn, C6, and C8O2. It might be more meaningful to use the corresponding central atoms and ligands from the standard simplification.

If a structure has several interpenetrating nets, such as the example in the screenshot, don’t use the structure for a start. If one wants to include it, the two nets must be considered separately. Hopefully, there won’t be many structures like that.

The descriptor of the net types are some strings. The coordination sequence and the point symbol of the net are probably the easiest to start with. The coordination sequence consists of the ten entries in the Cum list. If there are different ones, perhaps one can use one behind the other.

The point symbol of the net and its VS (vertex symbol) can be found at the very end of each entry (before the line with hashes). In the example below, the point symbol is 5^5.8 and the vertex symbol is 5.5.5.5.5.8(4).

Point symbol for net: {5^5.8}
4-c net; uninodal net

Topological type: unj; 4/5/h3 (topos&RCSR.ttd) {5^5.8} - VS [5.5.5.5.5.8(4)] (17963 types in 4 databases)
##